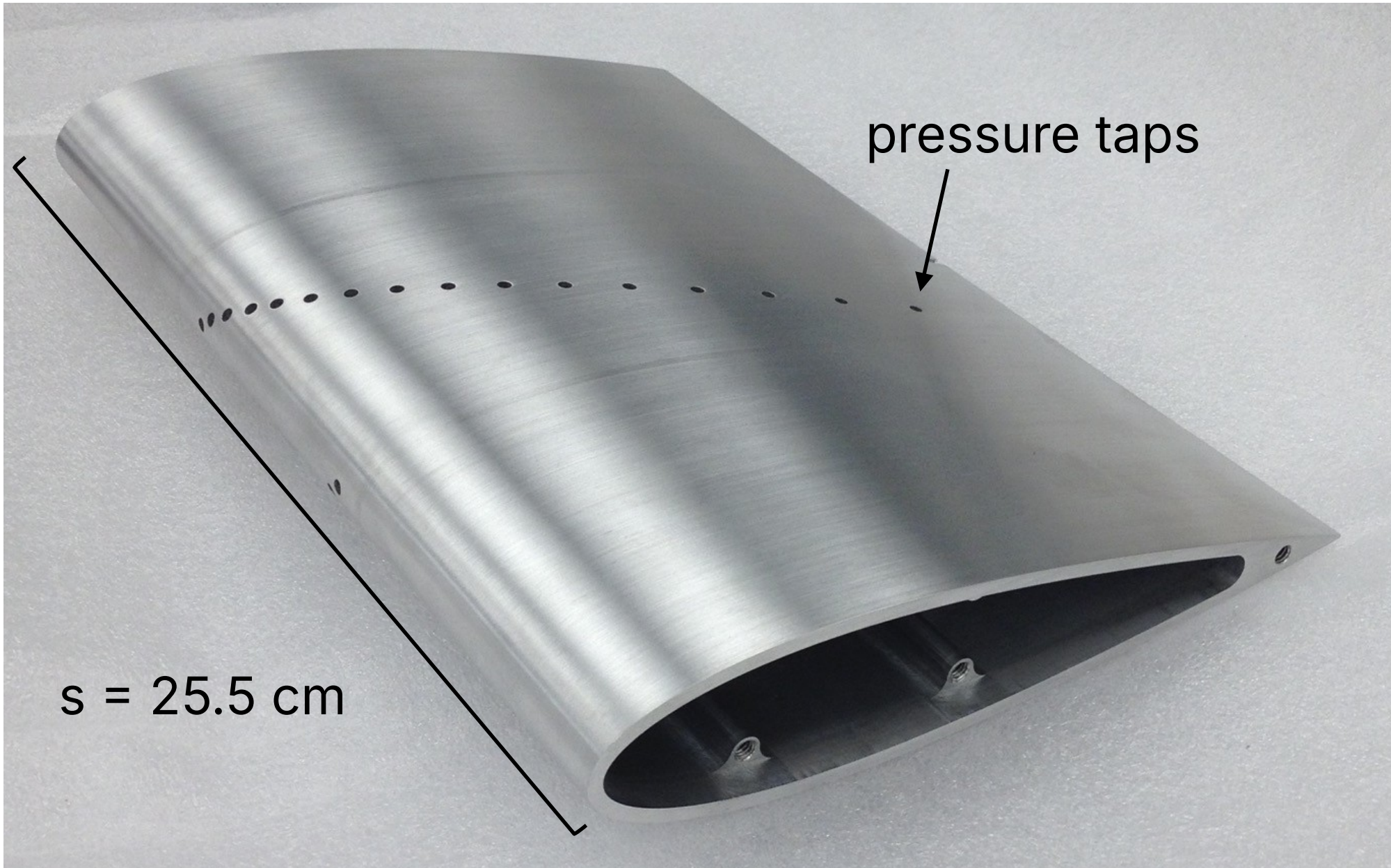
Experimental Setup

Airfoil Design

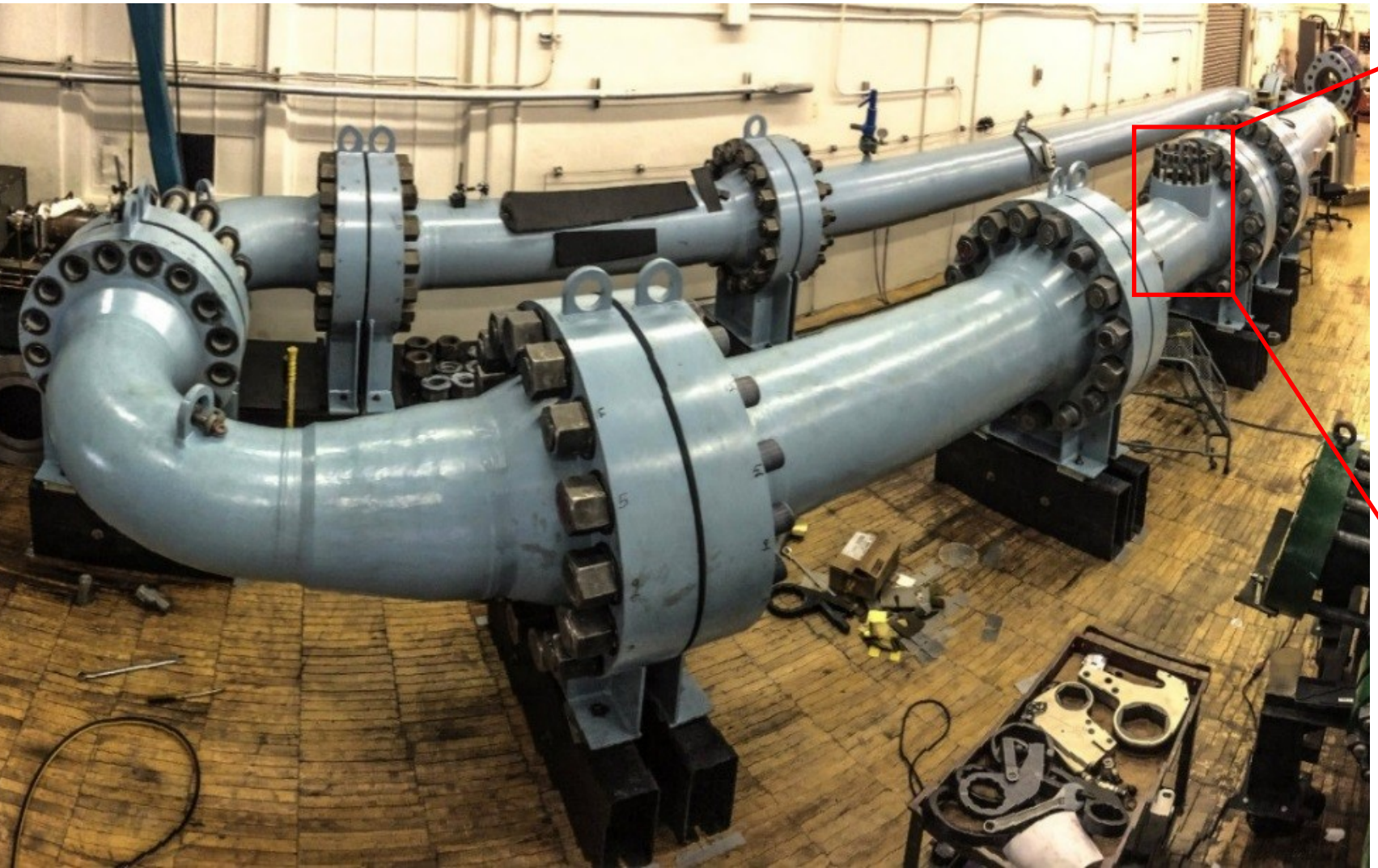


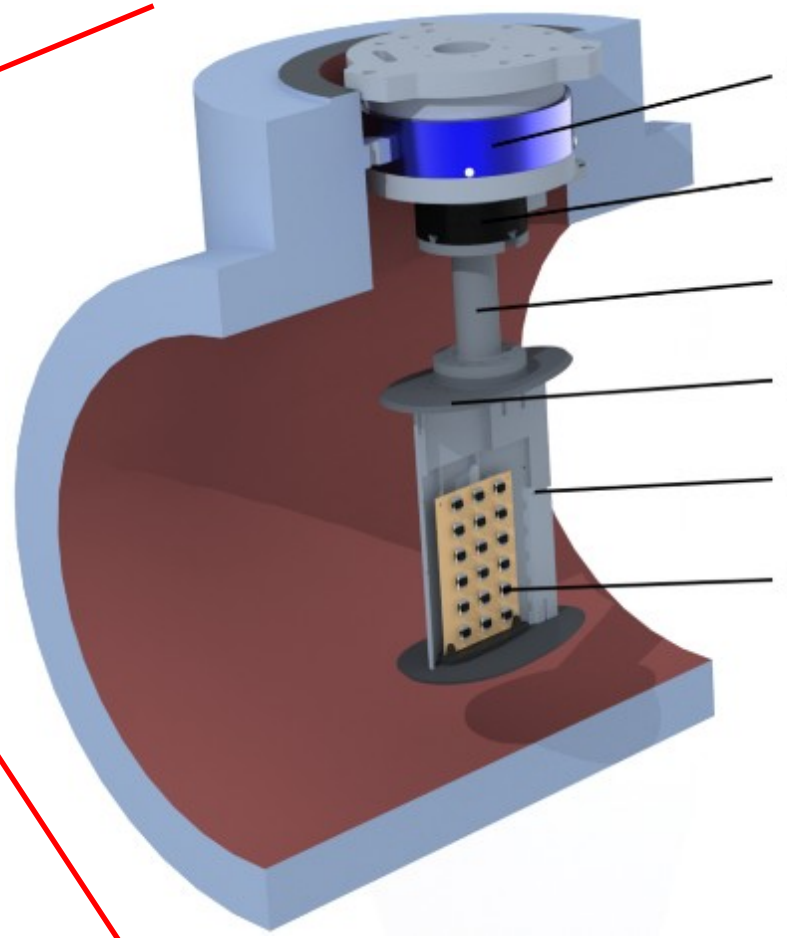


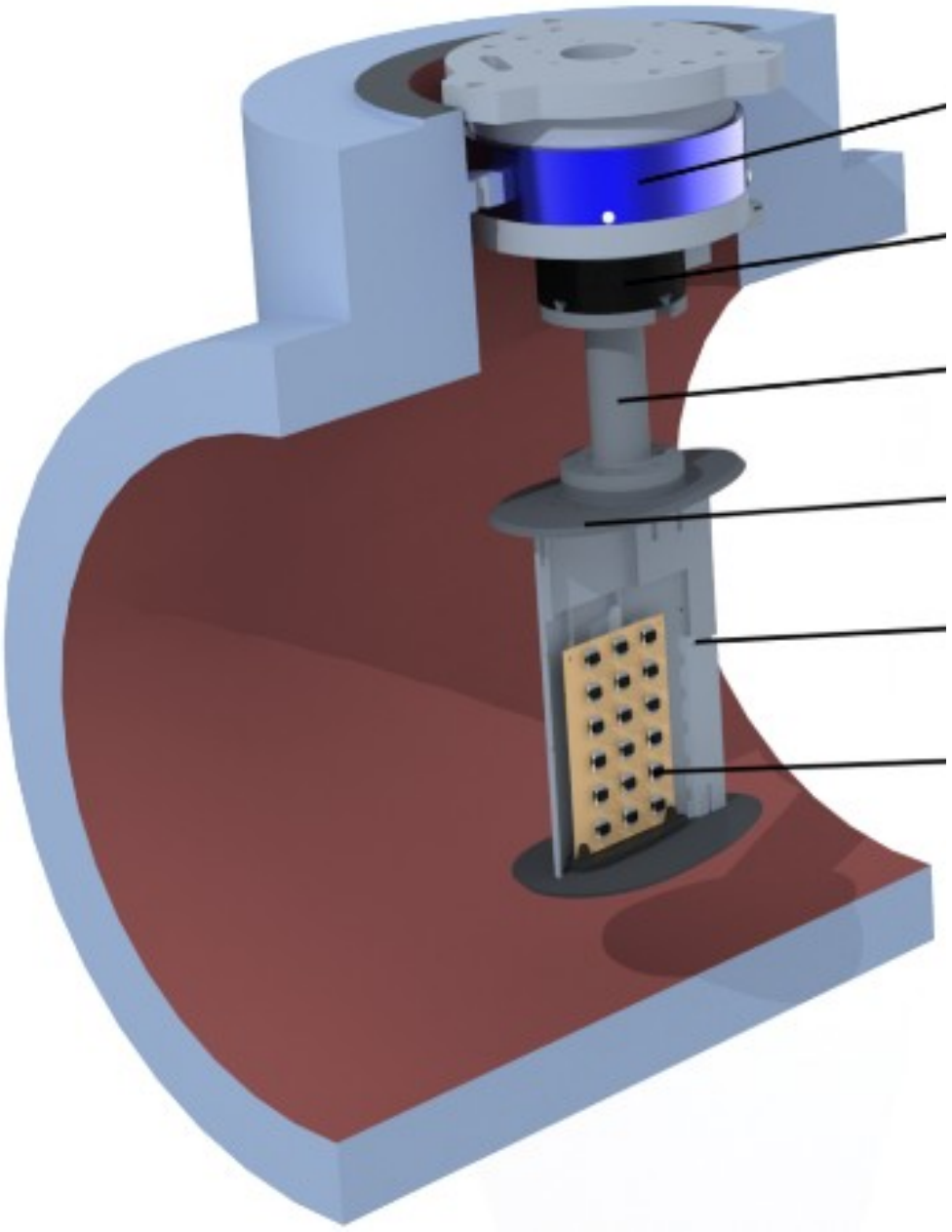
Using Pressure to Achieve High Reynolds Numbers

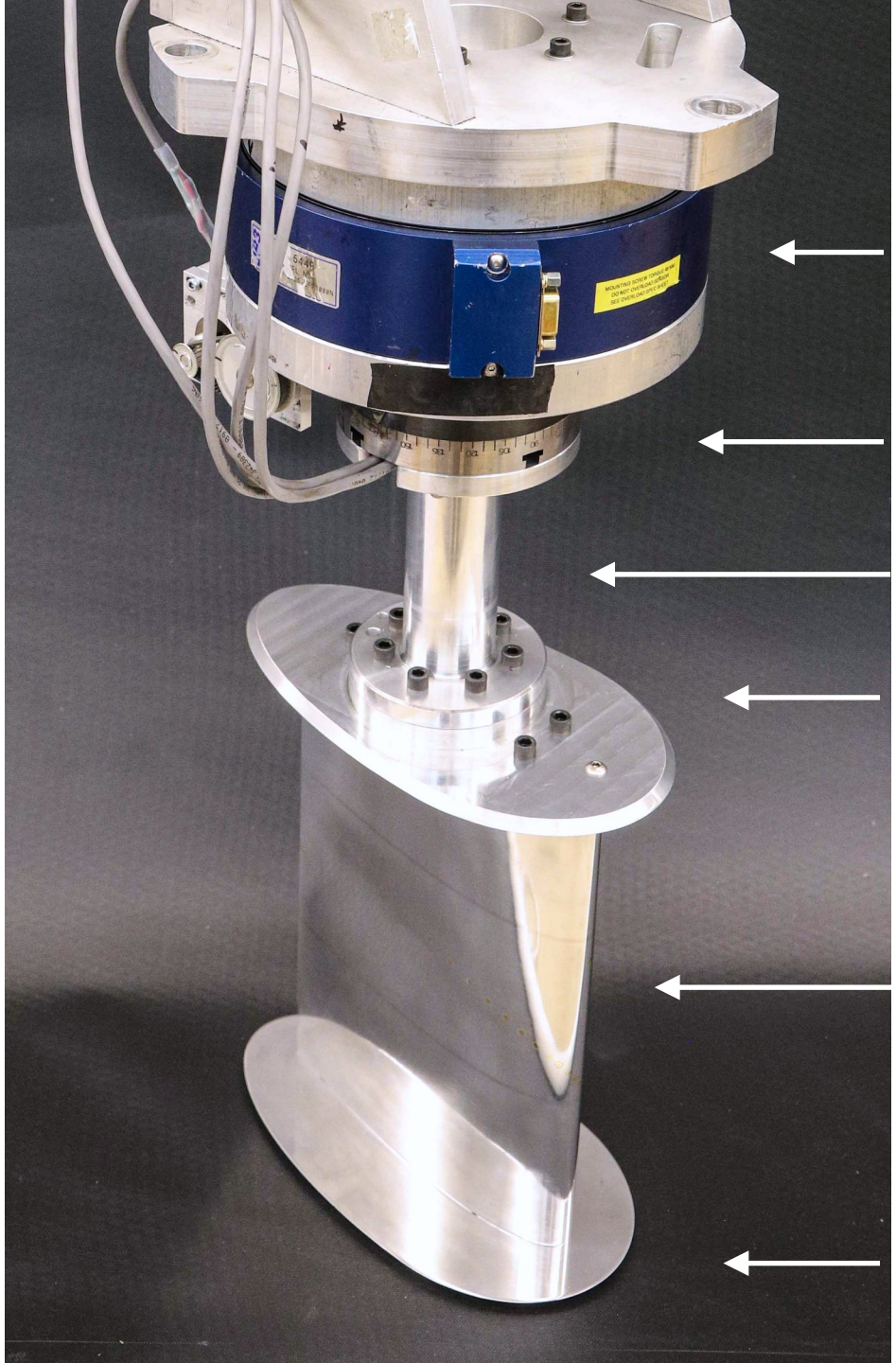
230 atm → 200 times higher Re

$$Re_c = \frac{\rho U_\infty C}{\mu}$$








force sensor
stepper motor
mounting rod
endplate
airfoil
endplate

Data Analysis

- Three data sets, each at a different Reynolds number. Calculate the Reynolds number of each dataset!
- Choose one Reynolds number and plot the pressure distribution of the top surface and the bottom surface at zero angle of attack. Is this what you would expect?
- Choose one Reynolds number and plot the pressure distribution on the top of the airfoil at the angle right before stall, and the angle right after stall. Compare them.

Calculating lift and drag

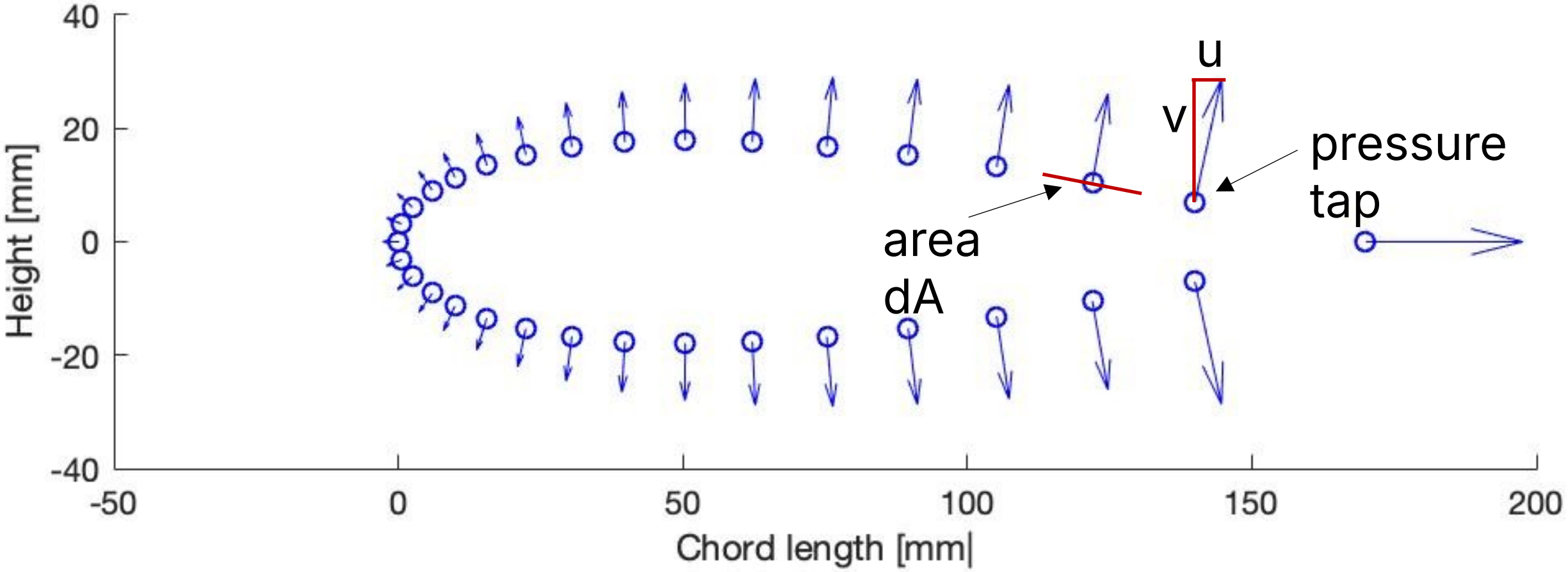
a) From force sensor:
it directly measures lift and drag on the contraption below.

b) From pressure sensors:
integrate pressure over the entire surface → resultant force on airfoil
find components normal (lift) and parallel (drag) to the flow

c) Compare! Do you expect them to be the same? Are they measuring
the exact same things?

Integrating pressure

```
[xn, yn, u, v] =
airfoil_normal(angle);
```



Data Analysis

- For each Reynolds number, plot the lift and drag curves based on the force sensor data and based on the pressure sensor data. Compare them.
- Make one plot that shows the lift and drag curves of the pressure sensors for all three Reynolds numbers. Compare them.